

Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

by

Md. Rakibul Hasan Mridha
ID: 2020333545

Md Sagor Hossain Saddas
ID: 2020333553

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DEPARTMENT OF CIVIL ENGINEERING

Supervised by

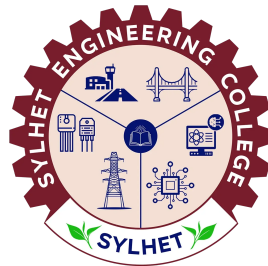
Md Johir Bin Alam
Professor

Department of CEE
Shahjalal University of Science & Technology

Co-Supervised by

Mohiuddin Arif
Lecturer

Department of Civil Engineering
Sylhet Engineering College



Sylhet Engineering College
(Shahjalal University of Science & Technology)

Sylhet-3100, Bangladesh

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Candidate's Declaration

This is to certify that the work presented in this thesis entitled, “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**”, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

Signature of the Candidates

Md. Rakibul Hasan Mridha
Reg. No.: 2020333545

Md Sagor Hossain Saddas
Reg. No.: 2020333553

Recommendation Letter from Thesis Supervisor

The thesis entitled “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**” submitted by the students:

1. Md. Rakibul Hasan Mridha (Reg. No.: 2020333545)
2. Md Sagor Hossain Saddas (Reg. No.: 2020333553)

is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor’s Degree.

Signature of the Supervisor

Md Johir Bin Alam

Professor

Department of CEE

Shahjalal University of Science & Technology

Signature of the Co-Supervisor

Mohiuddin Arif

Lecturer

Department of Civil Engineering

Sylhet Engineering College

Date: _____

Certificate of Acceptance

The thesis titled “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**” submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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Sylhet Engineering College

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Department of Civil Engineering
Sylhet Engineering College

External Examiner

Md Johir Bin Alam
Professor
Department of CEE
Shahjalal University of Science & Technology

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The Authors

Abstract

Keywords:

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1 Introduction

1.1 Background and Motivation

Bangladesh's road network expansion relies heavily on bitumen for flexible pavements, yet quality variations between local and imported sources profoundly affect long-term performance. Imported low-cost bitumen, often sourced from Iran, has frequently faced bans by agencies such as the Roads and Highways Department (RHD) due to rapid pavement deterioration and early rutting. In contrast, Eastern Refinery's laboratory-grade bitumen maintains a stable pricing structure at approximately Tk6,300 per unit while offering superior stability. Despite this, a comprehensive empirical evaluation comparing these variants under Bangladesh's subtropical climate is essential to optimize construction costs without compromising safety and durability. Furthermore, the structural integrity of the pavement is significantly dictated by aggregate properties. Incorporating varying shapes of coarse aggregates, specifically replacing conventional aggregates with a proportion of cubical aggregates, has been shown to improve the mechanical response of bituminous mixtures, necessitating an integrated study of both bitumen source and aggregate shape.

1.2 Problem Statement

Existing studies generally focus on fundamental bitumen properties or waste modifiers, but lack direct, localized comparisons between Eastern Refinery laboratory-grade bitumen and imported low-cost bitumen. Moreover, no comprehensive analysis integrates economic factors with essential performance metrics like load-bearing capacity and climate resistance in high-traffic areas. Additionally, while the importance of aggregate shape is acknowledged, systematic investigations analyzing the synergistic effects of locally available bitumen and specifically proportioned cubical aggregates on pavement performance remain scarce. This gap hinders policy-making for sustainable, cost-effective, and durable road construction practices in Bangladesh.

1.3 Research Objectives

To assess and compare the performance and cost-effectiveness of local laboratory bitumen versus imported low-cost bitumen for Bangladesh road applications, focusing on the influence of aggregate shape. The specific objectives include:

1. To determine and compare the fundamental properties (penetration, ductility, viscosity, specific gravity, and softening point) of laboratory bitumen and imported bitumen.
2. To investigate the effect of substituting 20% of the coarse aggregate fraction (19 mm to 12.5 mm) with cubical aggregates on the structural integrity of the bituminous mix.
3. To evaluate pavement performance metrics, including Marshall stability, flow, indirect tensile strength (ITS), and climate resistance (modified Marshall immersion).
4. To perform a cost-benefit analysis integrating initial material costs with projected maintenance economics to determine long-term viability.

1.4 Significance of the Study

Optimizing bitumen selection combined with strategic aggregate shape modifications can significantly reduce maintenance costs and enhance the road economy in Bangladesh, where infrastructure supports rapid urbanization. Superior local bitumen paired with optimal mix designs improves pavement lifespan, minimizing disruptions and the environmental impact of frequent repairs. The results of this study will inform RHD specifications, promote self-reliance in construction materials, and guide future sustainable infrastructure development.

- 2 Literature Review**
- 3 Methodology**
- 4 Results and Discussion**
- 5 Conclusion**

References